claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-a3456c131daf47453.jsonl`
- SHA-256: `5013d0b924e883f81d4795d8d6d4001533dbfc21d4fcc5e03e0f21d4f69fc8c0`
- Source modified: `2026-06-06T17:35:07+00:00`
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- project: `wf_6237b110-d58`
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`

Transcript

1. user (2026-06-06T17:33:55.173Z)

Adversarial Claim Verifier (voter 1/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):
- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:
1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ?

especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio).

Claim under review
"An ML classifier on acoustic waveforms (XGBoost) detects tennis ball-racket impacts with median per-fold accuracy 0.98 (range 0.96-0.99), demonstrating audio-based hit detection is highly accurate in isolation."

Source: https://pmc.ncbi.nlm.nih.gov/articles/PMC11843912/ (primary)
Supporting quote: "accuracy across each fold ranged between 0.96 and 0.99, with a median value of 0.98"

Checklist
1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

refuted=true if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.
refuted=false ONLY if: claim is well-supported, current, and source quality matches claim strength.
Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-06T17:33:59.468Z)

I'll verify this claim about XGBoost achieving 0.98 median accuracy for tennis ball-racket impact detection. Let me fetch the primary source.

3. user (2026-06-06T17:34:12.494Z)

XGBoost Classifier Performance for Tennis Impact Detection

(1) XGBoost Accuracy for Impact Detection

The XGBoost model achieved high performance metrics:

> "In the five-fold cross-validation, the accuracy across each fold ranged between 0.96 and 0.99, with a median value of 0.98"

This represents the first model, specifically trained for distinguishing impacts from background noise.

(2) Classification Task

The system performs **binary classification of impact vs. noise**, not shot-type or continuous temporal detection:

> "The first model was intended to distinguish between impact and sharp background noise"

The algorithm detects acoustic peaks in audio and classifies each peak as either a legitimate ball-impact sound or environmental noise.

(3) Dataset Composition

The training dataset comprised **1,400 audio samples**:

> "The algorithm was trained based on a dataset containing 1400 audio samples...we gathered 509 sounds for the class noise, 442 for rebound, and 448 for the impact."

These appear to be **segmented/isolated sound events** rather than continuous match recordings used for training.

(4) Cross-Validation Setup

The study employed **5-fold cross-validation**:

> "each fold containing 1120 samples for training and 280 for testing"

(5) Limitations and Class Balance

Class imbalance existed: The noise class (509 samples) outnumbered rebound samples (442). The study notes that rebounds presented greater classification difficulty, requiring an ensemble method rather than XGBoost alone. Additionally, the system's on-court accuracy (85%) was substantially lower than laboratory ML accuracy (95%+), suggesting environmental complexity wasn't fully represented in training data.

4. assistant (2026-06-06T17:34:16.456Z)

This is highly informative. Let me note the key issue: the 0.98 median accuracy is for a binary classification task (impact vs. noise on pre-segmented peaks), but real-world on-court accuracy drops to 85%. Let me dig deeper to confirm the "in isolation" framing and the real-world drop.

5. user (2026-06-06T17:34:26.038Z)

Analysis of System Accuracy Claims

(1) On-Court vs. Lab Accuracy: The 85% Figure

The study states: "the average accuracy of the whole system that was applied on-court was 85%." This figure appears in the abstract but lacks detailed breakdown in the results section showing how this differs from controlled conditions.

For groundstrokes specifically, on-court detection rates were: "96.25% of cases in G1 and 81.25% in G2" for impacts, and "86.25% of cases in G1 and 70.83% in G2" for rebounds.

(2) Second Model Purpose

The ensemble was designed for **rebound detection**: "The second model proved more difficult to implement because it had to identify the less harsh rebounds." The ensemble combined XGBoost, SVM, and MLP using soft voting for classifying rebounds versus noise.

(3) Accuracy on 2-Class vs. 3-Class Problem

The 0.98 accuracy was **2-class only**. The first model distinguished "between noise and impact" (binary). The second model (0.97 median accuracy) was also binary: "differentiating between the rebound and noise."

There was no unified 3-class classifier; instead: "Two models were created to implement the classification of sound events."

(4) Precision and Recall

The paper reports **only accuracy and Cohen's kappa**, not precision/recall separately. No confusion matrices or per-class metrics are provided.

(5) Separate Peak-Detection Step and Its Recall

Yes, there was a prior peak-detection stage. For groundstrokes: "In G1, 98 peaks with impact shape, 262 with rebound features, and 99.640 other sounds" were detected. The ML then filtered these. However, **recall of the peak-detection step is not explicitly reported**?only downstream ML classification accuracy.

6. assistant (2026-06-06T17:34:30.971Z)

I now have a clear picture. Let me assess the claim against the evidence.

The claim states: "An ML classifier on acoustic waveforms (XGBoost) detects tennis ball-racket impacts with median per-fold accuracy 0.98 (range 0.96-0.99), demonstrating audio-based hit detection is highly accurate in isolation."

The literal numbers (0.98 median, 0.96-0.99 range) are correctly quoted from the paper. The phrase "in isolation" is actually a fair qualifier ? the 0.98 IS the isolated-classification accuracy, and the full pipeline drops to 85% on-court. Let me do one search to check if there's any contradicting evidence or if these numbers are anomalous/disputed.

7. user (2026-06-06T17:34:42.074Z)

Web search results for query: "tennis racket impact sound detection machine learning accuracy audio classification"

Links: [{"title":"Tennis Timing Assessment by a Machine Learning-Based Acoustic Detection System: A Pilot Study - PMC","url":"https://pmc.ncbi.nlm.nih.gov/articles/PMC11843912/"},{"title":"Tennis Timing Assessment by a Machine Learning-Based Acoustic Detection System: A Pilot Study | MDPI","url":"https://www.mdpi.com/2411-5142/10/1/47"},{"title":"(PDF) Ball Hit Detection in Table Tennis Games Based on Audio Analysis","url":"https://www.researchgate.net/publication/224649455_Ball_Hit_Detection_in_Table_Tennis_Games_Based_on_Audio_Analysis"},{"title":"Tennis Timing Assessment by a Machine Learning-Based Acoustic Detection System: A Pilot Study","url":"https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11843912/"},{"title":"Spin Detection Using Racket Bounce Sounds in Table Tennis This research was funded by Sony AI.","url":"https://arxiv.org/html/2409.11760v1"},{"title":"Machine Learning Classifiers for Speech Detection | Request PDF","url":"https://www.researchgate.net/publication/361771281_Machine_Learning_Classifiers_for_Speech_Detection"},{"title":"Sound-Based Spin Estimation in Table Tennis: Dataset and Real-Time Classification Pipeline This work was funded by Sony AI","url":"https://arxiv.org/html/2409.11760"},{"title":"[2409.11760] Sound-Based Spin Estimation in Table Tennis: Dataset and Real-Time Classification Pipeline","url":"https://arxiv.org/abs/2409.11760"},{"title":"Spin Detection Using Racket Bounce Sounds in Table Tennis","url":"https://arxiv.org/pdf/2409.11760v1"}]

Based on the search results, here's what I found regarding tennis racket impact sound detection using machine learning:

Detection Accuracy

A machine learning algorithm for detecting tennis racket impact sounds showed detection accuracy higher than 95%, while the average accuracy of the whole system applied on-court was 85%. Another neural network approach using a CNN with MFCC and delta coefficients achieved classification accuracy of over 90% for every class.

System Methodology

The acoustic detection system classifies the sound of the ball's impact on the racket and the ground to measure the time between them and give immediate feedback to the player. The dataset included 509 sounds for noise, 442 for rebound, and 448 for the impact of the ball hitting the racket strings.

Advanced Applications

Recent research has extended beyond simple impact detection. Detected sound segments are converted to Mel spectrograms and classified using CNNs to identify the bounce surface and detect whether spin was applied. Spin type in tennis could be predicted from sound with over 70% accuracy.

Practical Applications

The system has proven effective in evaluating the technical skills of players on the court and highlighting their areas for improvement, showing significant potential for practical applications in player training and performance analysis.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. assistant (2026-06-06T17:34:52.358Z)

The search corroborates the source. The paper is the primary source (a peer-reviewed MDPI/PMC pilot study), and the numbers are accurately quoted. Let me now assess the claim against the checklist.

The claim has two parts:
1. **The factual numbers**: "median per-fold accuracy 0.98 (range 0.96-0.99)" ? these are EXACTLY quoted and verified from the primary source.
2. **The interpretation**: "demonstrating audio-based hit detection is highly accurate in isolation" ? the word "in isolation" is actually load-bearing and correct here.

Key nuances I verified:
- The 0.98 is a **binary classification** (impact vs. noise) accuracy on **pre-segmented audio peaks** (1,400 isolated samples), NOT end-to-end detection on continuous match audio.
- It is a **pilot study** with a modest dataset.
- The **full on-court pipeline accuracy drops to 85%** (and as low as 70.83% for rebounds in some groups) ? this is precisely why the qualifier "in isolation" matters and is appropriate.
- Precision/recall and peak-detection recall are NOT reported, so the isolated classifier's 0.98 doesn't translate to end-to-end hit-detection performance.

The claim's wording is careful: it explicitly says "in isolation," which correctly distinguishes the classifier-stage accuracy from the full-system performance. The numbers are not overreached, not contradicted, not outdated (2025 publication), and not marketing. The source is a peer-reviewed primary study matching the claim's strength.

The one weakness: a reader could over-extrapolate this to "audio hit detection works great end-to-end," but the claim itself carefully avoids that by saying "in isolation." Since the claim is faithful to the source and self-qualifies appropriately, I cannot in good faith refute it ? the numbers and the qualifier both hold up.

9. user (2026-06-06T17:35:02.702Z)

Structured output provided successfully

10. assistant (2026-06-06T17:35:07.874Z)

Verification complete. The claim survives adversarial review (refuted=false, high confidence).

The numbers (0.98 median, 0.96-0.99 range) are quoted verbatim from the primary peer-reviewed source, and the qualifier "in isolation" is the critical reason it holds up: the 0.98 is the standalone XGBoost classifier accuracy on pre-segmented audio peaks (binary impact-vs-noise, 1,400 isolated samples, 5-fold CV), not end-to-end detection. The same paper's full on-court pipeline drops to 85%, which the "in isolation" wording correctly distinguishes rather than overreaching. Because the claim is faithful to the source and self-qualifies appropriately, it cannot be refuted ? though it remains a single small pilot study, so any downstream use should not extrapolate to end-to-end recall.